Algorithm Sample 8

Dataset-expansion sample

Procedure

The pseudocode below is drawn from a source-backed image-to-LaTeX benchmark and reproduced verbatim.

Algorithm 1 Delta Particle Filter

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1. **Input:** data $y_{1:T}$, level $l \in \mathbb{N}$, particle number $N \in \mathbb{N}$ and parameter $\theta \in \Theta$.
2. **Initialize:** For $i \in \{1, ..., N\}$, independently generate $\overline{W}^{i,l}_{\Delta_l:1}$ from $\mathcal{N}(0, \Delta_l)$. For $(i, k) \in \{1, ..., N\} \times \{1, ..., \Delta_{l-1}^{-1}\}$ set

$$\overline{W}^{i,l-1}_{k\Delta_{l-1}} = \overline{W}^{i,l}_{2k\Delta_l} + \overline{W}^{i,l}_{(2k-1)\Delta_l}.$$

   Set $t = 1$, $\hat{\tilde{p}}^N(y_{1:0}) = 1$ for convention and go to step 3.

3. **Iterate:** For $i \in \{1, ..., N\}$ compute

$$\tilde{u}^i_t = \frac{\max\Big\{\kappa_{t,l}\Big(\overline{w}^{i,l}_{\Delta_l:t}\Big), \kappa_{t,l-1}\Big(\overline{w}^{i,l-1}_{\Delta_{l-1}:t}\Big)\Big\}}{\sum_{j=1}^N \max\Big\{\kappa_{t,l}\Big(\overline{w}^{j,l}_{\Delta_l:t}\Big), \kappa_{t,l-1}\Big(\overline{w}^{j,l-1}_{\Delta_{l-1}:t}\Big)\Big\}}.$$

   Set $\hat{p}^N(y_{1:t}) = \hat{p}^N(y_{1:t-1}) \frac{1}{N} \sum_{i=1}^N \max\Big\{\kappa_{t,l}\Big(\overline{w}^{i,l}_{\Delta_l:t}\Big), \kappa_{t,l-1}\Big(\overline{w}^{i,l-1}_{\Delta_{l-1}:t}\Big)\Big\}$. Then sample $\Big(\overline{w}^{1:N,l}_{\Delta_l:t}, \overline{w}^{1:N,l-1}_{\Delta_{l-1}:t}\Big)$ with replacement from $\Big(\overline{w}^{1:N,l}_{\Delta_l:t}, \overline{w}^{1:N,l-1}_{\Delta_{l-1}:t}\Big)$ using probabilities $\tilde{u}^{1:N}_t$. For $i \in \{1, ..., N\}$, independently generate $\overline{W}^i_{t+\Delta_l:t+1}$ from $\mathcal{N}(0, \Delta_l)$. For $(i, k) \in \{1, ..., N\} \times \{1, ..., \Delta_{l-1}^{-1}\}$ set

$$\overline{W}^{i,l-1}_{t+k\Delta_{l-1}} = \overline{W}^{i,l}_{t+2k\Delta_l} + \overline{W}^{i,l}_{t+(2k-1)\Delta_l}.$$

   Set $t = t + 1$ and if $t = T$ go to step 4, otherwise restart step 3.

4. **Grand Selection:** For $i \in \{1, ..., N\}$ compute

$$\tilde{u}^i_T = \frac{\max\Big\{\kappa_{T,l}\Big(\overline{w}^{i,l}_{\Delta_l:T}\Big), \kappa_{T,l-1}\Big(\overline{w}^{i,l-1}_{\Delta_{l-1}:T}\Big)\Big\}}{\sum_{j=1}^N \max\Big\{\kappa_{T,l}\Big(\overline{w}^{j,l}_{\Delta_l:T}\Big), \kappa_{T,l-1}\Big(\overline{w}^{j,l-1}_{\Delta_{l-1}:T}\Big)\Big\}}.$$

   Set $\hat{p}^N(y_{1:T}) = \hat{p}^N(y_{1:T-1}) \frac{1}{N} \sum_{i=1}^N \max\Big\{\kappa_{T,l}\Big(\overline{w}^{i,l}_{\Delta_l:T}\Big), \kappa_{T,l-1}\Big(\overline{w}^{i,l-1}_{\Delta_{l-1}:T}\Big)\Big\}$. Sample one $\Big(\overline{w}^{l}_{\Delta_l:T}, \overline{w}^{l-1}_{\Delta_{l-1}:T}\Big)$ from $\Big(\overline{w}^{1:N,l}_{\Delta_l:T}, \overline{w}^{1:N,l-1}_{\Delta_{l-1}:T}\Big)$ using $\tilde{u}^{1:N}_T$ and go to step 5.

5. **Output:** trajectories $\Big(\overline{w}^{l}_{\Delta_l:T}, \overline{w}^{l-1}_{\Delta_{l-1}:T}\Big)$ and normalizing constant estimate $\hat{\tilde{p}}^N(y_{1:T})$.